In [5]:
```python
import cv2  # Pragya Singhal, Cu24220050,43
import numpy as np

print("Lucas-Kanade Sparse Optical Flow...")
cap = cv2.VideoCapture("3987752-hd_1920_1080_24fps.mp4")

feature_params = dict(maxCorners=100, qualityLevel=0.3, minDistance=7, blockSize=7)

lk_params = dict(
    winSize=(15, 15),
    maxLevel=2,
    criteria=(cv2.TERM_CRITERIA_EPS | cv2.TERM_CRITERIA_COUNT, 10, 0.03),
)

color = np.random.randint(0, 255, (100, 3))
ret, old_frame = cap.read()
if ret:
  # Resize to prevent lag
  old_frame = cv2.resize(old_frame, (640, 360))
  old_gray = cv2.cvtColor(old_frame, cv2.COLOR_BGR2GRAY)
  p0 = cv2.goodFeaturesToTrack(old_gray, mask=None, **feature_params)
  mask = np.zeros_like(old_frame)

  while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
      break
    frame = cv2.resize(frame, (640, 360))
    frame_gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    p1, st, err = cv2.calcOpticalFlowPyrLK(
        old_gray, frame_gray, p0, None, **lk_params
    )
    if p1 is not None:
      good_new = p1[st == 1]
      good_old = p0[st == 1]

      # Draw the tracks    #Pragya Singhal, Cu24220050,43
      for i, (new, old) in enumerate(zip(good_new, good_old)):
        a, b = map(int, new.ravel())
        c, d = map(int, old.ravel())
        mask = cv2.line(mask, (a, b), (c, d), color[i].tolist(), 2)
        frame = cv2.circle(frame, (a, b), 5, color[i].tolist(), -1)

      img = cv2.add(frame, mask)
      cv2.imshow("Sparse Optical Flow (Lucas-Kanade)", img)

      old_gray = frame_gray.copy()
      p0 = good_new.reshape(-1, 1, 2)

    if cv2.waitKey(30) & 0xFF == ord("q"):
      break

cap.release()  # Pragya Singhal, Cu24220050,43
cv2.destroyAllWindows()

print("Starting Part 2: Farneback Dense Optical Flow...")
cap = cv2.VideoCapture("3987752-hd_1920_1080_24fps.mp4")
ret, frame1 = cap.read()

if ret:
  frame1 = cv2.resize(frame1, (640, 360))
  prvs = cv2.cvtColor(frame1, cv2.COLOR_BGR2GRAY)
  hsv = np.zeros_like(frame1)
  hsv[..., 1] = 255  # Set saturation to maximum

  while cap.isOpened():  # Pragya Singhal, Cu24220050,43
    ret, frame2 = cap.read()
    if not ret:
      break
    frame2 = cv2.resize(frame2, (640, 360))
    next_frame = cv2.cvtColor(frame2, cv2.COLOR_BGR2GRAY)

    flow = cv2.calcOpticalFlowFarneback(
        prvs, next_frame, None, 0.5, 3, 15, 3, 5, 1.2, 0
    )

    mag, ang = cv2.cartToPolar(flow[..., 0], flow[..., 1])
    hsv[..., 0] = ang * 180 / np.pi / 2  # Direction mapped to Hue
    hsv[..., 2] = cv2.normalize(mag, None, 0, 255, cv2.NORM_MINMAX)
    bgr = cv2.cvtColor(hsv, cv2.COLOR_HSV2BGR)

    cv2.imshow("Dense Optical Flow (Farneback)", bgr)
    prvs = next_frame

    if cv2.waitKey(30) & 0xFF == ord("q"):
      break

cap.release()  # Pragya Singhal, Cu24220050,43
cv2.destroyAllWindows()
print("Experiment execution completed.")
```

```
Lucas-Kanade Sparse Optical Flow...
Starting Part 2: Farneback Dense Optical Flow...
Experiment execution completed.
```